SPARQS: Silicon Photonic Architecture of Robust Quantum Sources for a scalable testbed in quantum computing

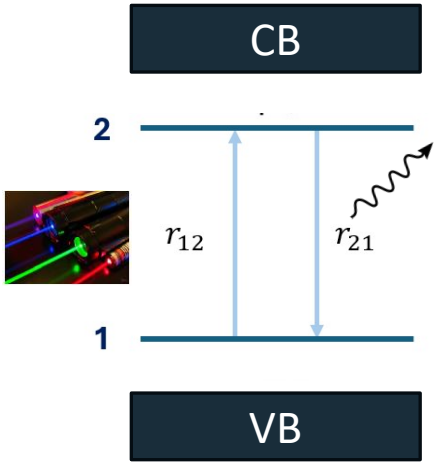
GRETA ANDRINI

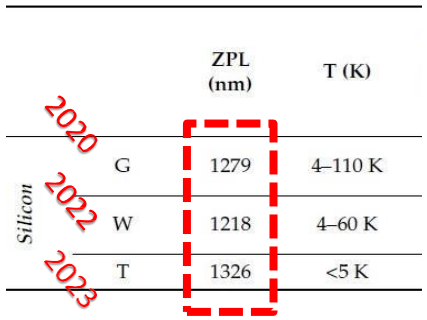
INFN – sez. di Torino

Single photon sources in silicon

State of the art

- Point defects: individual transitions→**on-demand** single photon emission with given properties
- Unique **spin-photon interface**





Andrini G. et al., Photonics 2024, 11(2)

- ☑ Bright **telecom wavelength single photon sources T<110K**
- ☑ Silicon-based solutions benefit from **microelectronics and integrated photonics know-how**

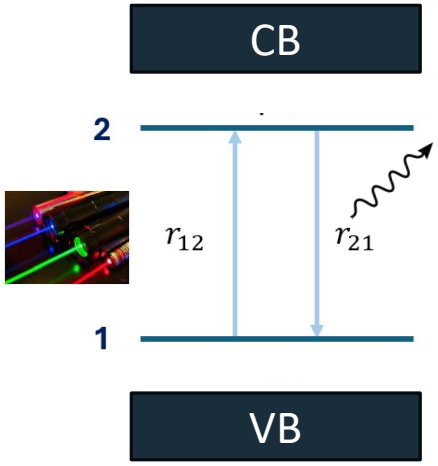
Applications:

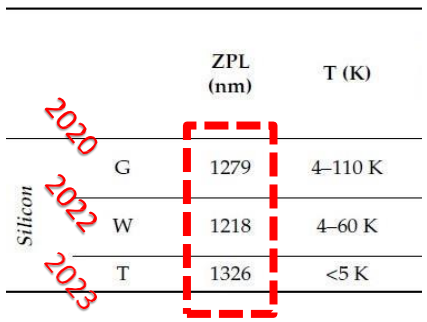
- Quantum communication
- Quantum information processing

Single photon sources in silicon

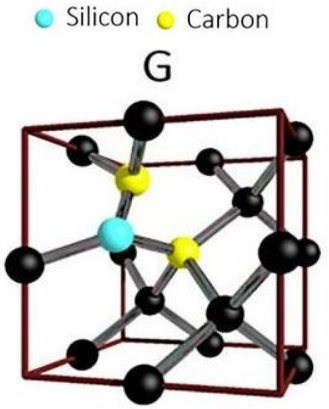
State of the art

- Point defects: individual transitions→**on-demand** single photon emission with given properties
- Unique **spin-photon interface**





Andrini G. et al., Photonics 2024, 11(2)

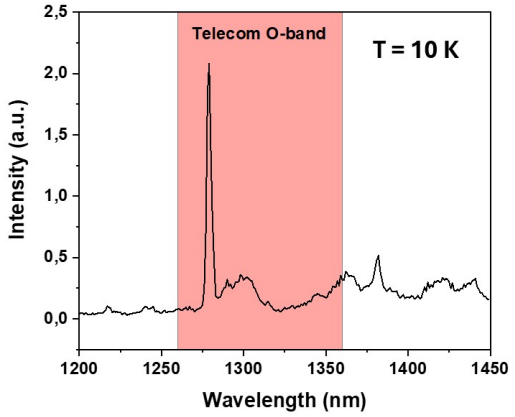


M. Khoury& M. Abbarchi, J. Appl. Phys. 131, 200901 (2022)

- ☑ Bright **telecom wavelength single photon sources T<110K**
- ☑ Silicon-based solutions benefit from **microelectronics and integrated photonics know-how**

Applications:

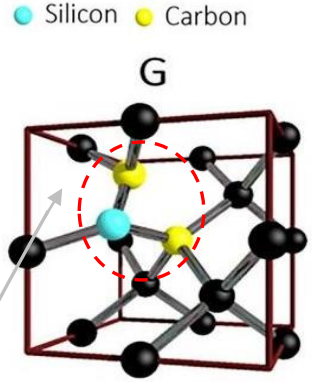
- Quantum communication
- Quantum information processing



Ensemble spectrum of G centers

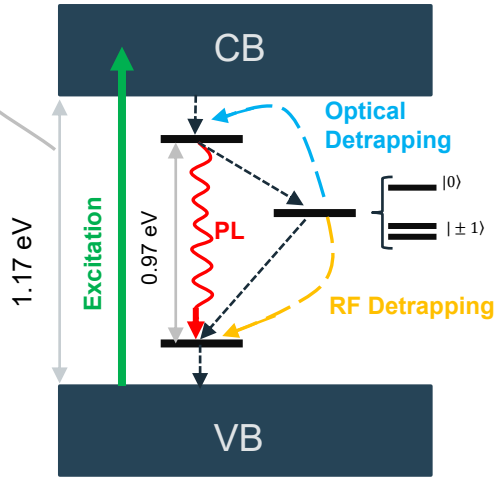
Single photon sources in silicon

State of the art



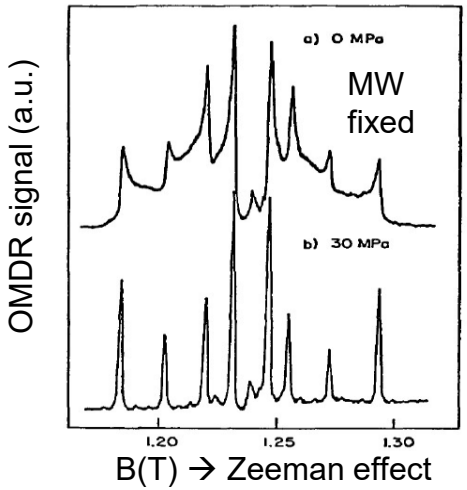
- RF field can control **spin-dependent radiation dynamics** of the metastable triplet state
- Optically Detected Magnetic Resonances are affected by nuclear spins → **quantum storage**

M. Khoury& M. Abbarchi, J. Appl. Phys. 131, 200901 (2022)



G center's energy levels

First evidence on ensemble (80s)

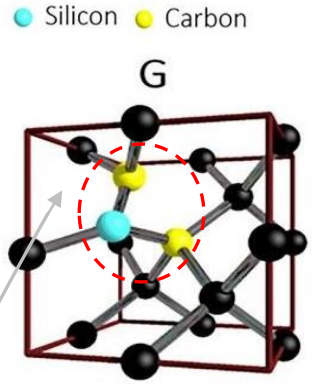


B(T) → Zeeman effect

K. M. Lee et al., Phys. Rev. Lett. 48, 37 (1982)

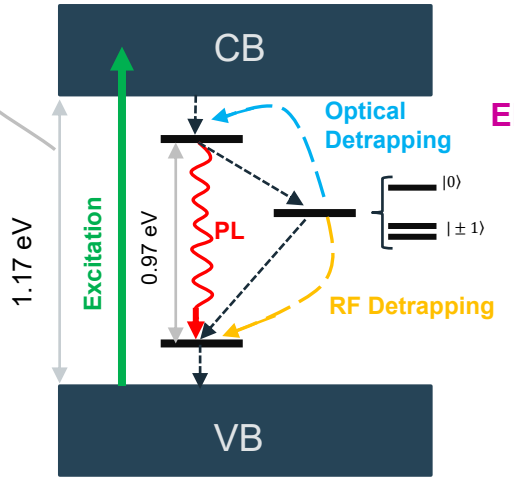
Single photon sources in silicon

State of the art

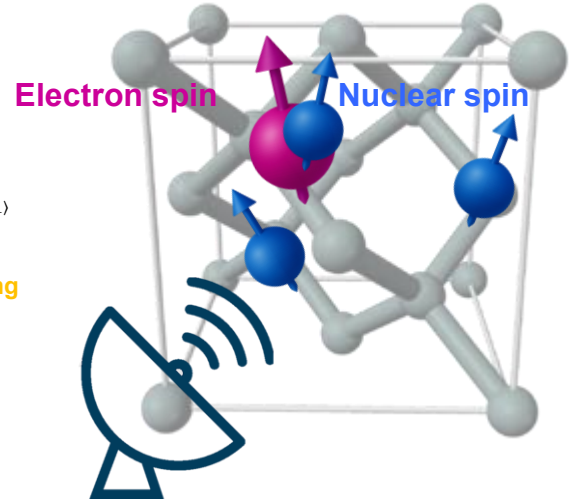


- ✓ RF field can control **spin-dependent radiation dynamics** of the metastable triplet state
- ? Optically Detected Magnetic Resonances are affected by nuclear spins → **quantum storage**

M. Khoury& M. Abbarchi, J. Appl. Phys. 131, 200901 (2022)



G center's energy levels

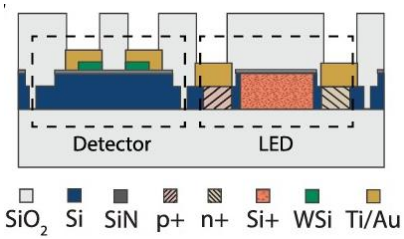


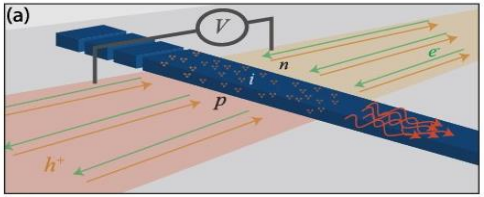
- **How** do the **nearby nuclear spins couple** to the G electronic spin?
- G center ground state S=0 → **long nuclear spins coherence time**

Silicon-based quantum optics

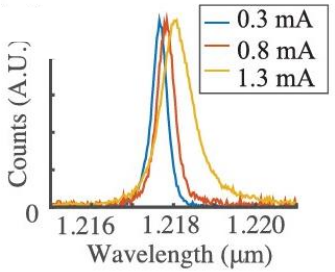
State of the art

- Electrically injected **monolithic solution** for cryogenic photonic integrated circuits with **ensembles** of W quantum emitters.





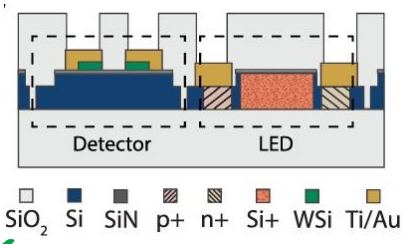
S. Buckley et al., Appl. Phys. Lett. 111, 141101 (2017)

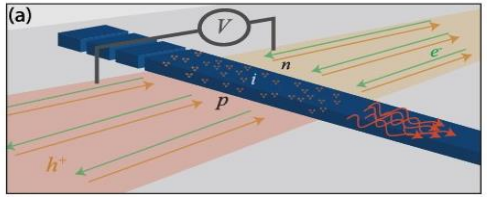


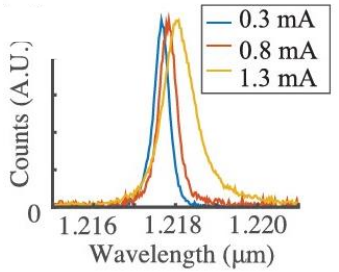
Silicon-based quantum optics

State of the art

- Electrically injected **monolithic solution** for cryogenic photonic integrated circuits with **ensembles** of W quantum emitters.

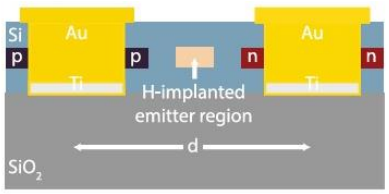


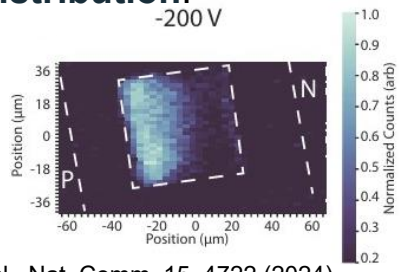


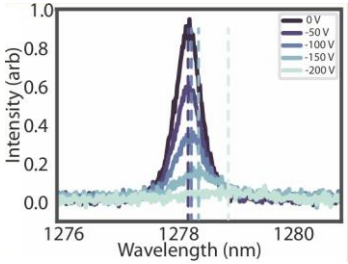


S. Buckley et al., Appl. Phys. Lett. 111, 141101 (2017)

- DC-field **manipulation** of **ensembles** of **G quantum emitters** embedded in a diode.
- Imaging of the **electric-field distribution**.





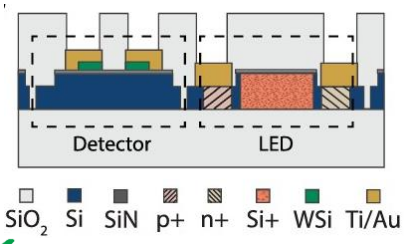


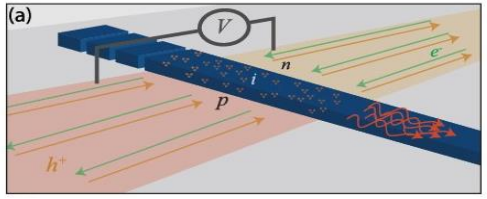
A. M. Day et al., Nat. Comm. 15, 4722 (2024)

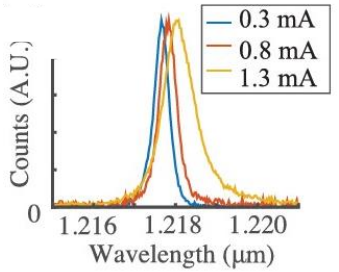
Silicon-based quantum optics

State of the art

- Electrically injected **monolithic solution** for cryogenic photonic integrated circuits with **ensembles** of W quantum emitters.

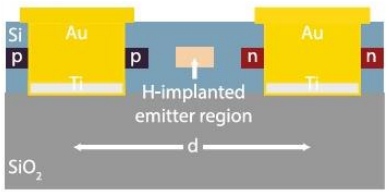


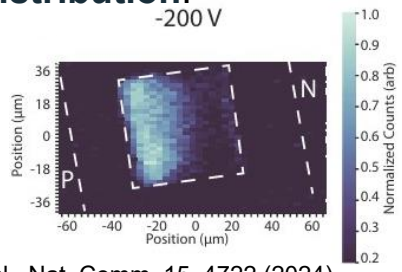


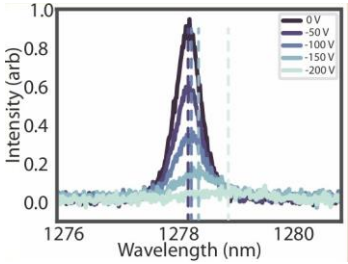


S. Buckley et al., Appl. Phys. Lett. 111, 141101 (2017)

- DC-field **manipulation** of **ensembles** of **G quantum emitters** embedded in a diode.
- Imaging of the **electric-field distribution**.

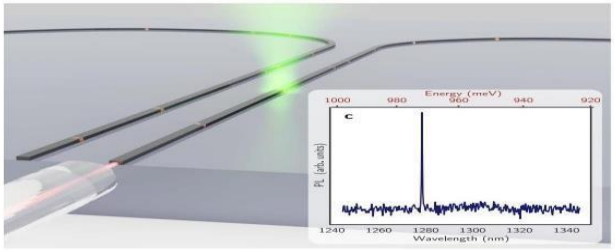




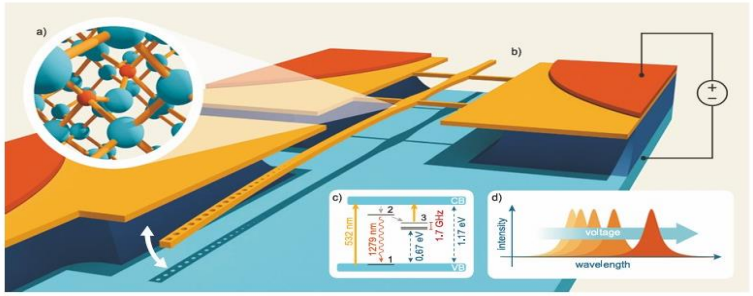


A. M. Day et al., Nat. Comm. 15, 4722 (2024)

- Tuning of **single G emitters'** spectral properties with **foundry-fabricated photonic integrated circuit (PIC)**.



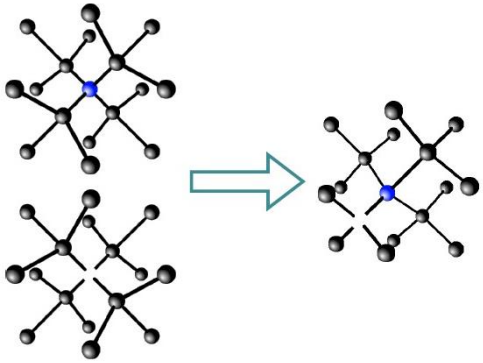
M. Prabhu et al., Nat. Comm. 14, 2380 (2023)

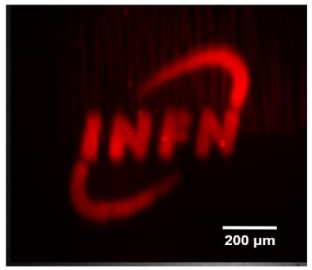


A. Buzzi et al., Nat. Comm. 16, 8829 (2025)

Creation of silicon quantum emitters

General rule: Ion Implantation + thermal treatment



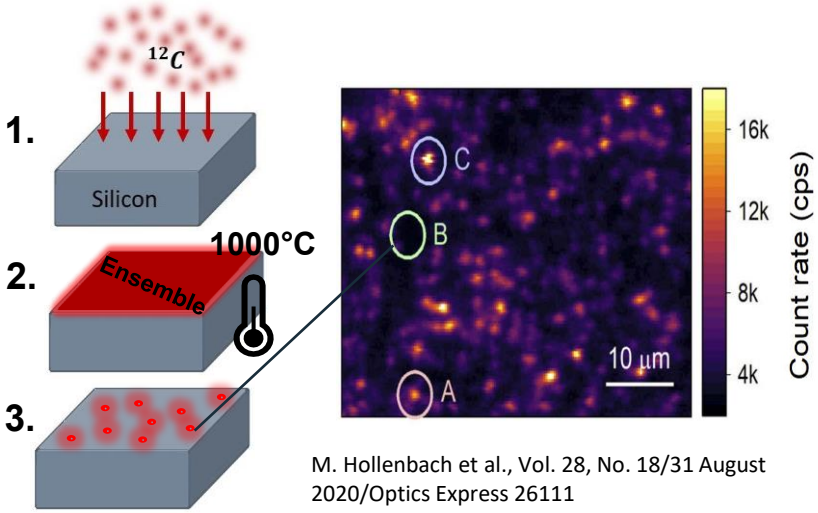


Ion Implantation at LNL INFN

☺ **Industrially available techniques**

☺ Controlled position and fluence

So far, isolating single emitters has mostly relied on a **purely statistical approach**



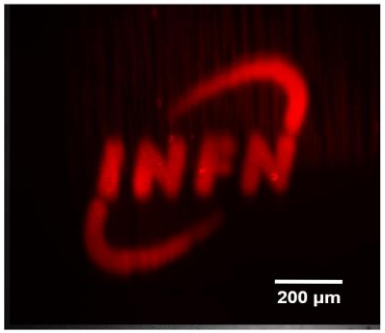
M. Hollenbach et al., Vol. 28, No. 18/31 August 2020/Optics Express 26111

☹ **Multi-step fabrication processes required** to integrate **randomly distributed** emitters into photonic structures.

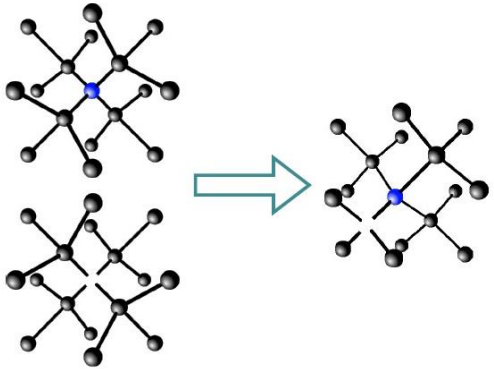
☹ Alternative Focused Ion Beam Technology can be used → still a **statistical approach**

Creation of silicon quantum emitters

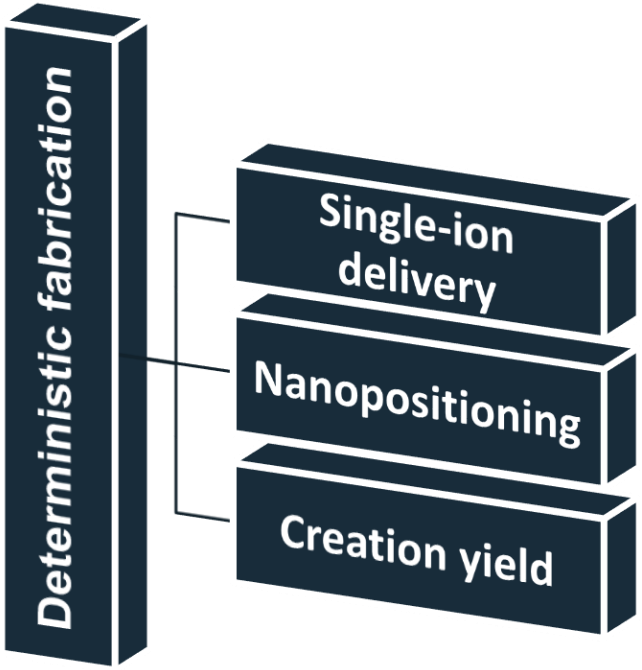
What are the main fabrication challenges?



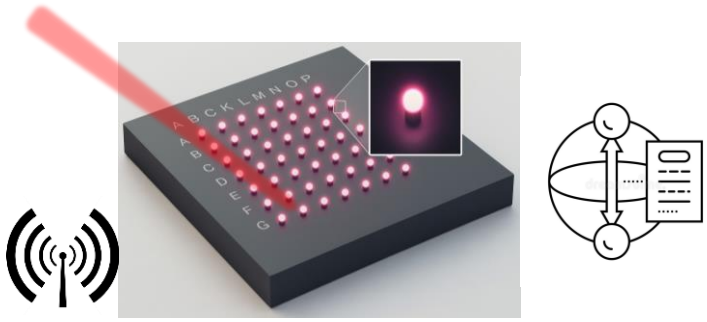
Ion Implantation at LNL INFN



Ion Implantation + thermal treatment



Project goal



<u>Goal 1:</u> platform-independent approach for the **controlled creation** of multiple **single-photon sources**

<u>Goal 2:</u> Realization of a **silicon qubit** platform for spin state **coherent manipulation**

Large-scale quantum networks and **quantum computing**

Project goal

Scalable, **silicon-based**, **fault-tolerant** **array of quantum emitters** that serves as a testbed for generating and **controlling photonic quantum states**.

How?

WP1: End-to-end architecture for wafer-scale production of telecom color centers in silicon
Coordinator: Greta Andrini, Postdoc, INFN-TO



Collimated ion beams technology and **in-situ laser processing**



Laser processing techniques for silicon defects engineering developed by the PI

Project goal

Scalable, **silicon-based**, **fault-tolerant** **array of quantum emitters** that serves as a testbed for generating and **controlling photonic quantum states**.

How?

WP2: Silicon qubit platform for spin state coherent manipulation
Coordinator: Gabriele Zanelli (Postdoc INRiM), Greta Andrini



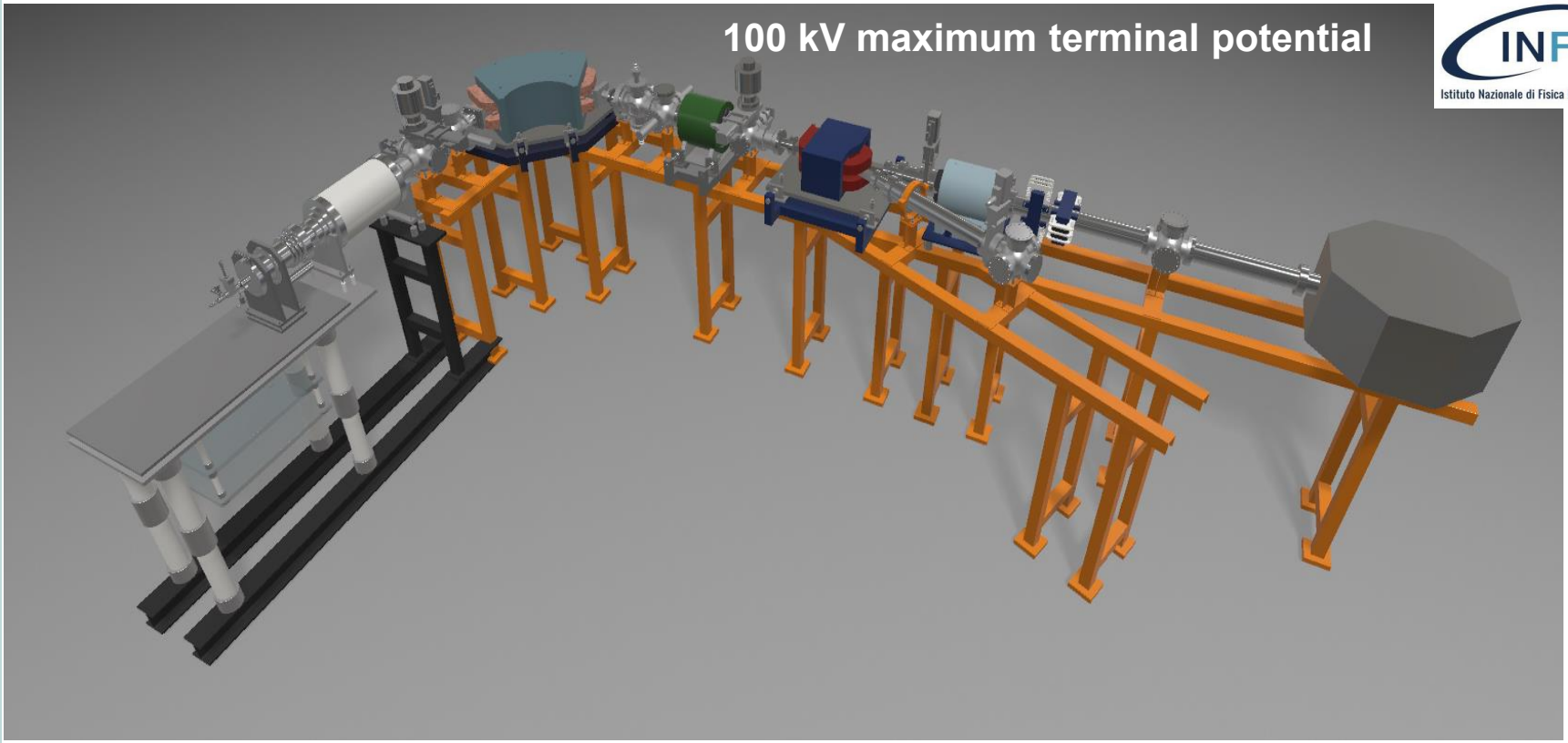
Post selection of **optimal emitters** through **cryogenic confocal microscopy** and **HBT interferometry**



Coherent control of qubits using the **Optically Detected Magnetic Resonance** technique.

Turin Ion Implantation facility

QUANTEP project INFN CSN5
Call "Quantum Technologies" 2020



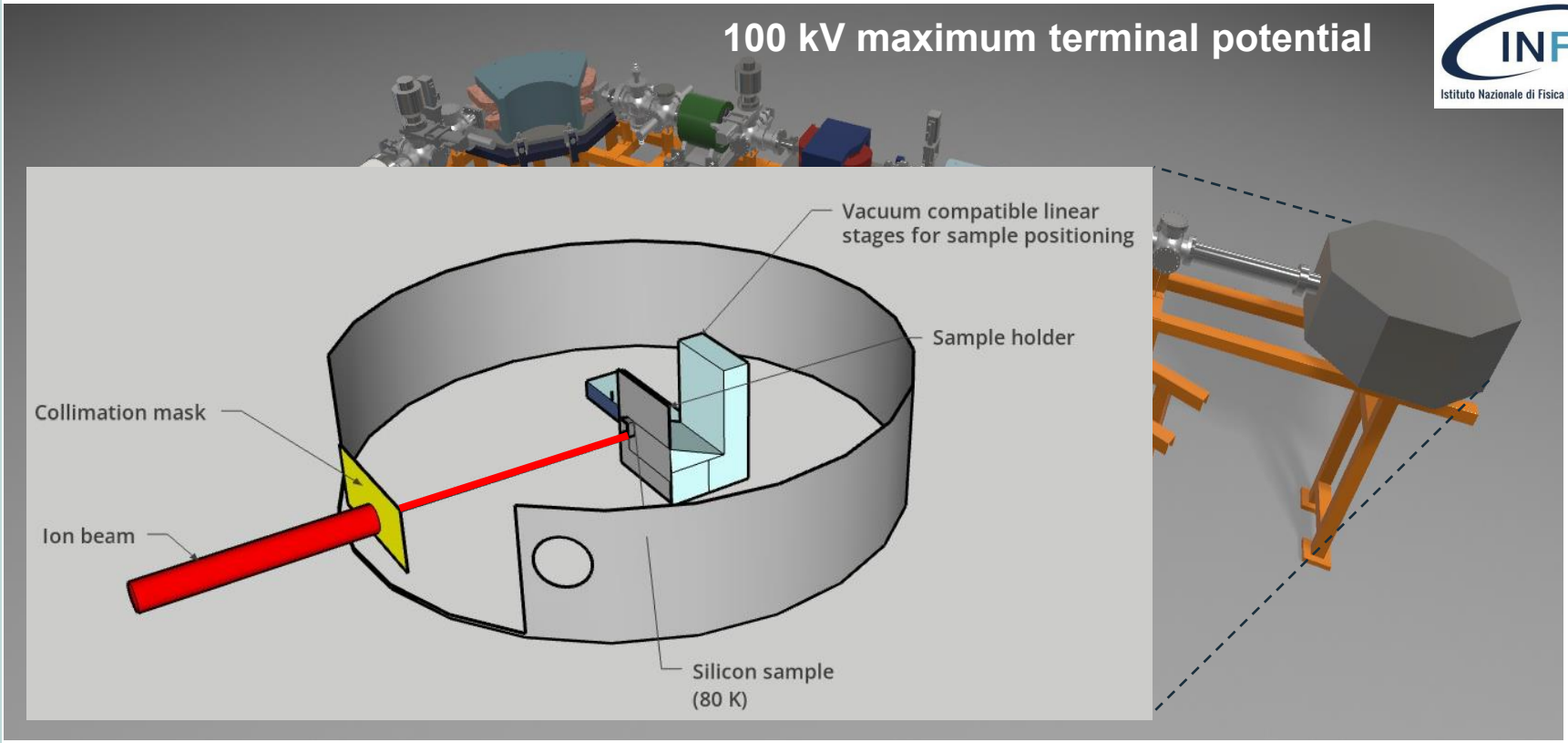
Laboratory for Ion Implantation at the University of Torino (LIUTo)

Turin Ion Implantation facility

DF ECCELLENZA
Università di Torino

QUANTEP project INFN CSN5
Call "Quantum Technologies" 2020

INFN
Istituto Nazionale di Fisica Nucleare

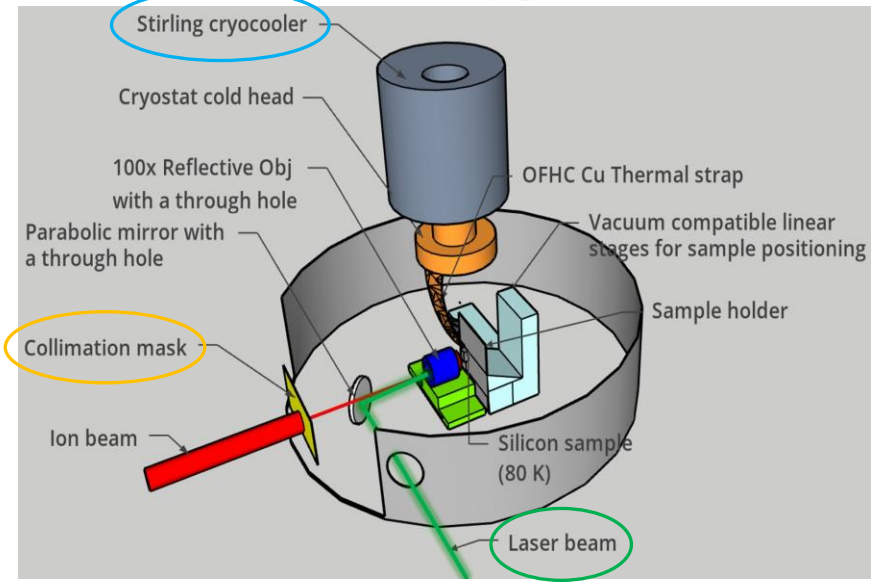


Laboratory for Ion Implantation at the University of Torino (LIUTo)

Proposal – WP 1

Ion implantation and in-situ laser annealing

Chamber upgrade

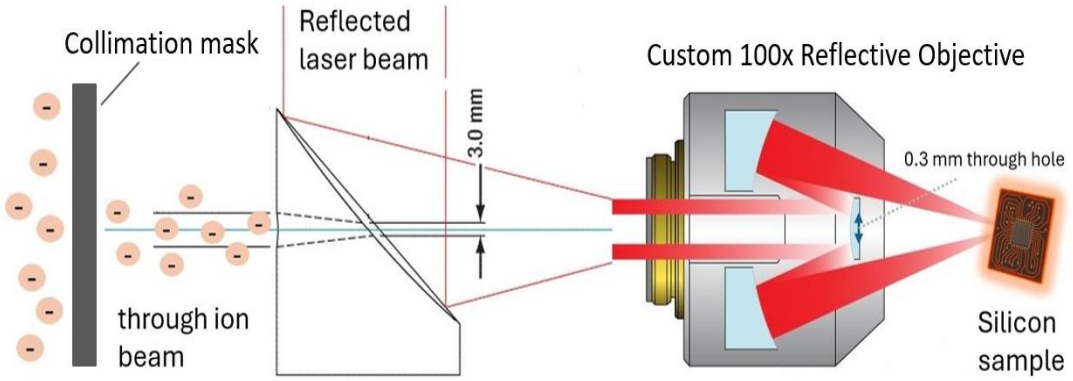


Device Fabrication

1. **In-situ** and **diffraction-limited** laser processing (100 µJ pulse)

2. Implantation resolution with **collimation masks** down to ~ 5µm

3. Stirling cryocooler for exploratory **in-situ confocal microscopy**



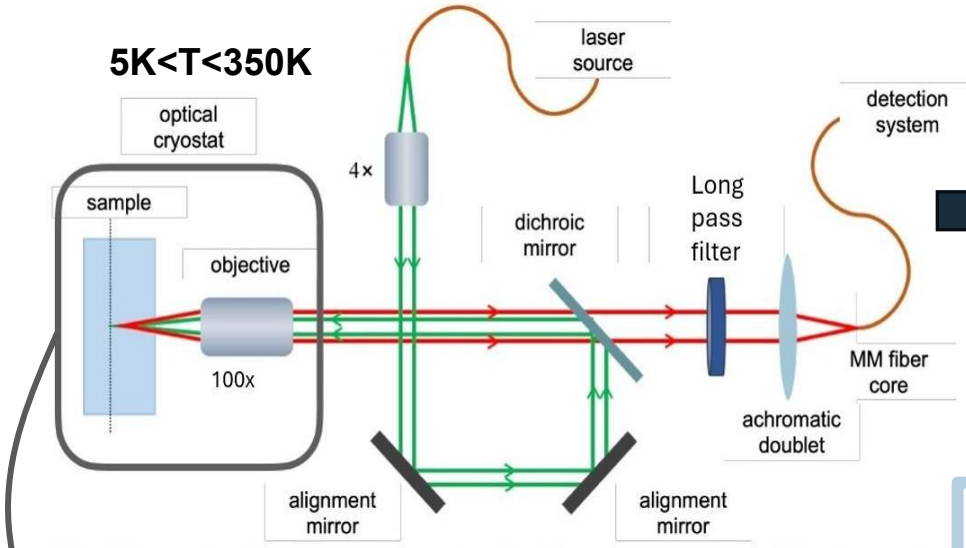
Custom Objective:
12 k€ (1° year)

Cryocooler: 42 k€ (2° year)

Optics and Optomechanics
(1° and 2° year): 31 k€

Characterization of the optical properites

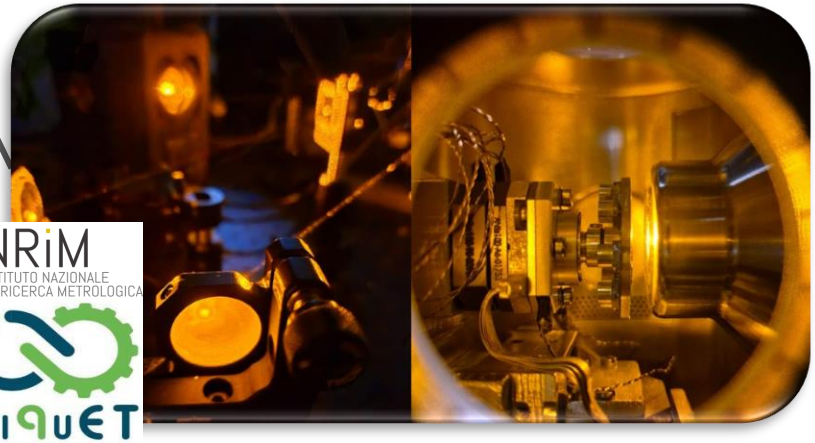
Telecom single-photon cryogenic confocal microscopy





Instrumentation at the Quantum Optics Laboratories – **Research Unit 2:**

1. **Superconductive Nanowire Single Photon Detector** (@1550 nm) T = 2.5 K
2. Interferometer for Fourier Transform **spectroscopy**



Device Characterization

- Optical sectioning → high contrast
- **Diffraction-limited resolution**

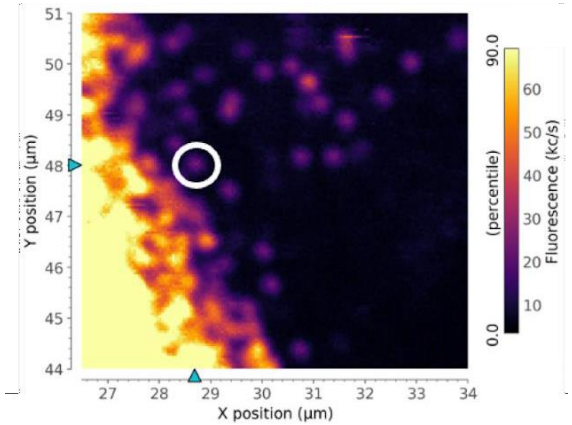
$$r_{x,y} = 0{,}6\frac{\lambda}{N.A.} \sim 1\ um$$

$$r_z = \frac{2n\lambda}{N.A.^2} \sim 4\ um$$

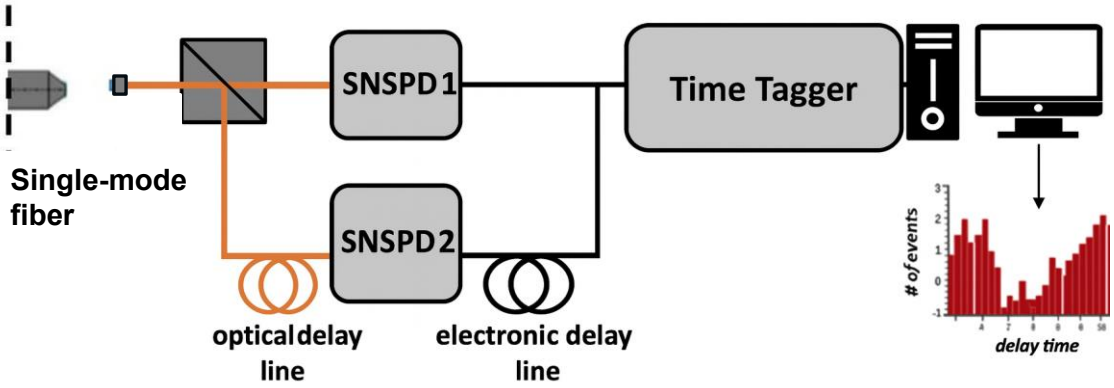
Coherent manipulation of silicon qubits

Step 1: Hanbury Brown & Twiss interferometry

Photoluminescence map



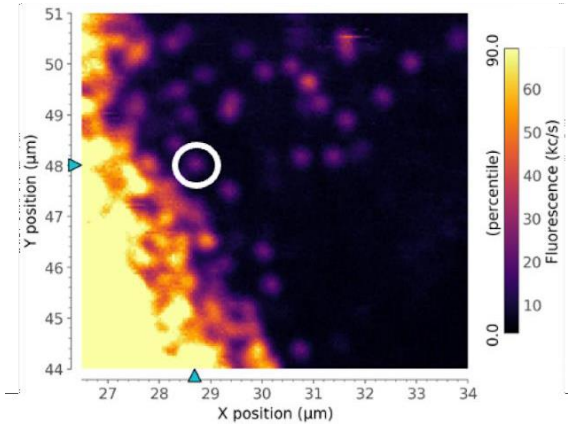
E. Corte & G. Andrini et al., ACS Photonics 2023, 10, 1, 101–110

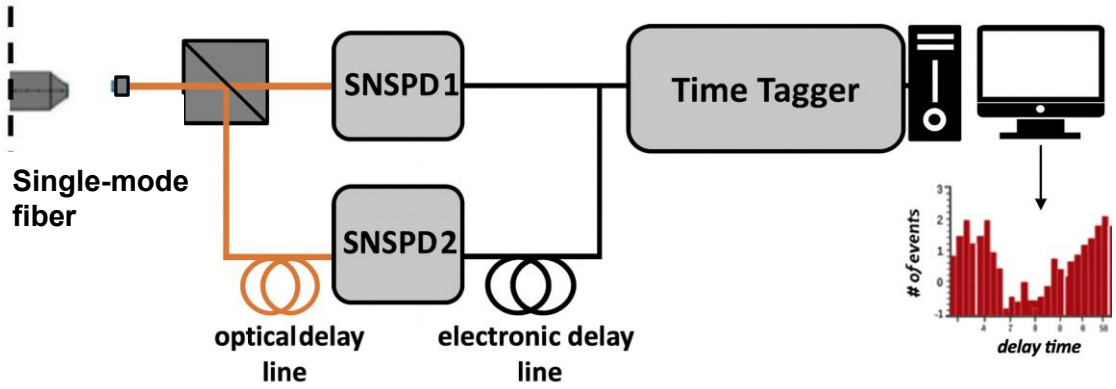


Coherent manipulation of silicon qubits

Step 1: Hanbury Brown & Twiss interferometry

Photoluminescence map





E. Corte & G. Andrini et al., ACS Photonics 2023, 10, 1, 101–110

Device Validation

Second-order autocorrelation function

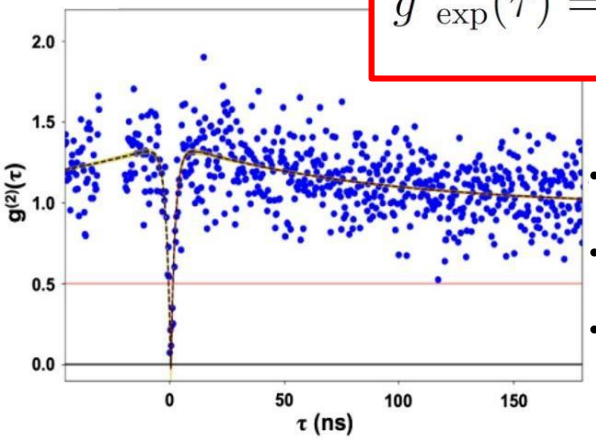
Classical expression

$$g^2(\tau) = \frac{\langle I(t) \cdot I(t+\tau) \rangle}{\langle I(t) \rangle^2}$$

Quantum formalism

$$\lim_{\tau \to 0} g^{(2)}(\tau) = g^{(2)}(0) \cong 1 - \frac{1}{n}$$

$$g^2{}_{\exp}(\tau) = \frac{c(t)}{R_1 R_2 w T}$$

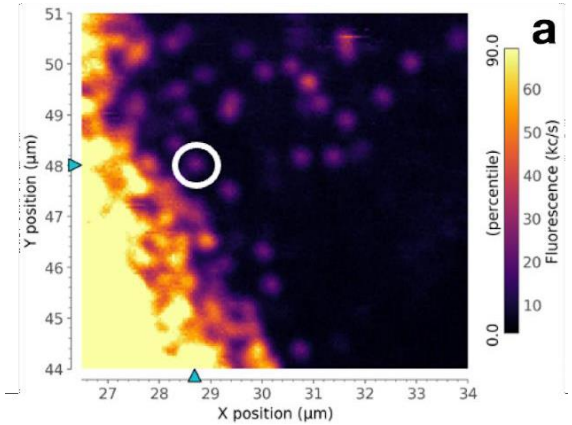


- R1, R2 = count rate detector 1,2;
- T = acquisition time;
- w = bin width

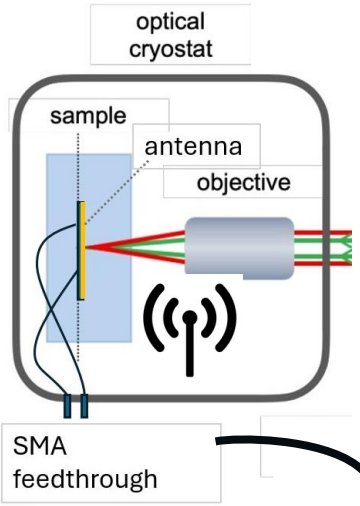
Coherent manipulation of silicon qubits

Step 2: CW Optically Detected Magnetic Resonance

Photoluminescence map



E. Corte & G. Andrini et al., ACS Photonics 2023, 10, 1, 101–110



Click signal



SNSPD 1

Multifunction I/O board

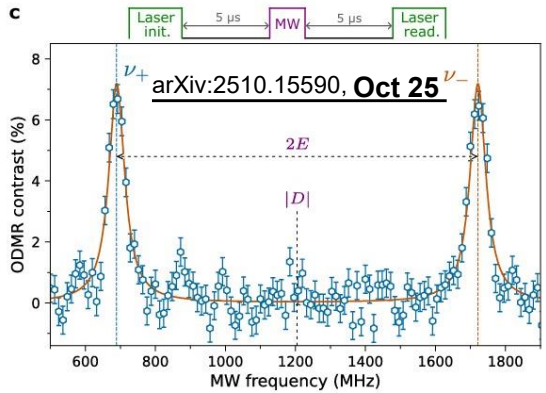
MW trigger

MW generator

Counts

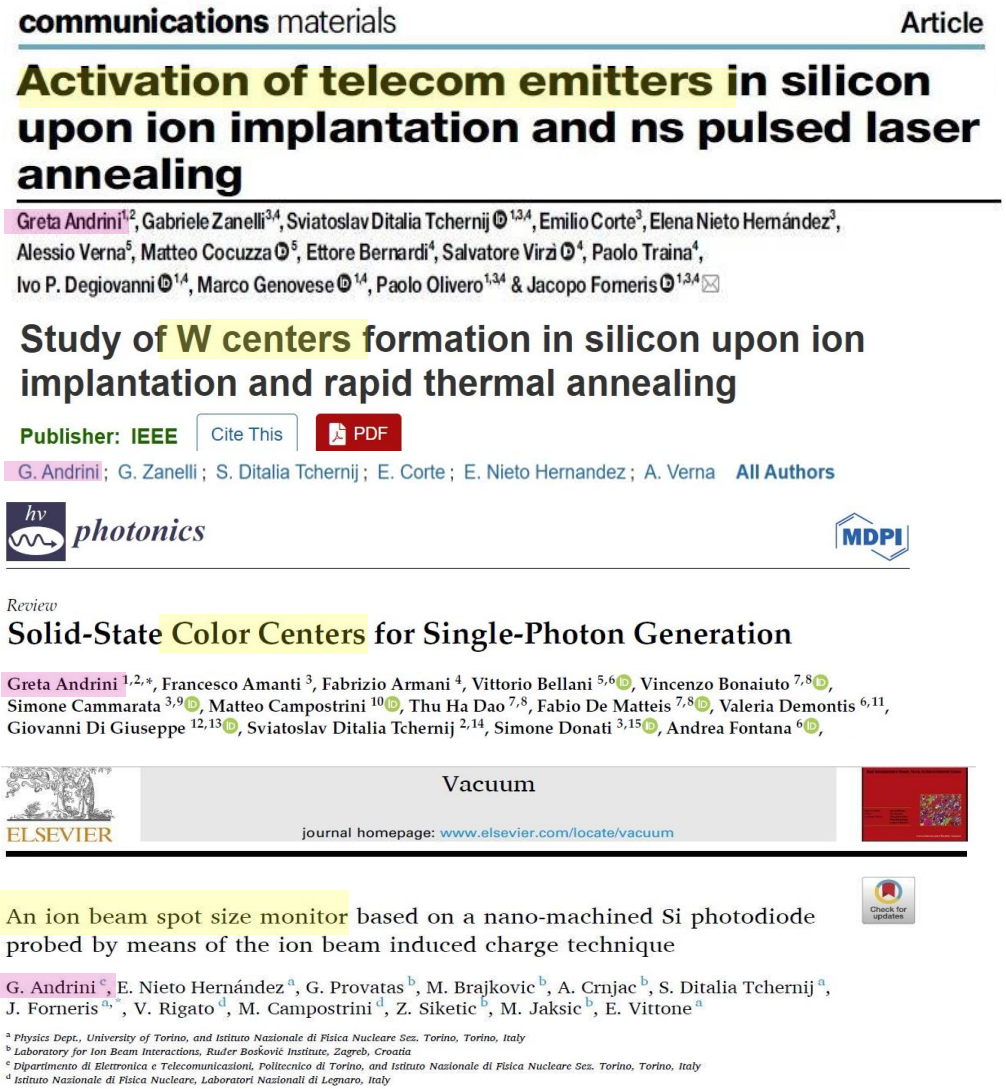
Counts

Device Validation



My expertises and SPARQS incomes

Supporting results



communications materials — Article

Activation of telecom emitters in silicon upon ion implantation and ns pulsed laser annealing

Greta Andrini[1,2], Gabriele Zanelli[3,4], Sviatoslav Ditalia Tchernij[1,3,4], Emilio Corte[3], Elena Nieto Hernández[3], Alessio Verna[5], Matteo Cocuzza[5], Ettore Bernardi[4], Salvatore Virzì[4], Paolo Traina[4], Ivo P. Degiovanni[1,4], Marco Genovese[1,4], Paolo Olivero[1,3,4] & Jacopo Forneris[1,3,4]

Study of W centers formation in silicon upon ion implantation and rapid thermal annealing

Publisher: IEEE | Cite This | PDF

G. Andrini ; G. Zanelli ; S. Ditalia Tchernij ; E. Corte ; E. Nieto Hernandez ; A. Verna All Authors

hv photonics — MDPI

Review

Solid-State Color Centers for Single-Photon Generation

Greta Andrini [1,2,*], Francesco Amanti [3], Fabrizio Armani [4], Vittorio Bellani [5,6], Vincenzo Bonaiuto [7,8], Simone Cammarata [3,9], Matteo Campostrini [10], Thu Ha Dao [7,8], Fabio De Matteis [7,8], Valeria Demontis [6,11], Giovanni Di Giuseppe [12,13], Sviatoslav Ditalia Tchernij [2,14], Simone Donati [3,15], Andrea Fontana [6],

ELSEVIER — Vacuum — journal homepage: www.elsevier.com/locate/vacuum

An ion beam spot size monitor based on a nano-machined Si photodiode probed by means of the ion beam induced charge technique

G. Andrini [c], E. Nieto Hernández [a], G. Provatas [b], M. Brajkovic [b], A. Crnjac [b], S. Ditalia Tchernij [a], J. Forneris [a,*], V. Rigato [d], M. Campostrini [d], Z. Siketic [b], M. Jaksic [b], E. Vittone [a]

[a] Physics Dept., University of Torino, and Istituto Nazionale di Fisica Nucleare Sez. Torino, Torino, Italy
[b] Laboratory for Ion Beam Interactions, Ruder Bošković Institute, Zagreb, Croatia
[c] Dipartimento di Elettronica e Telecomunicazioni, Politecnico di Torino, and Istituto Nazionale di Fisica Nucleare Sez. Torino, Torino, Italy
[d] Istituto Nazionale di Fisica Nucleare, Laboratori Nazionali di Legnaro, Italy

2 years from PhD, 16 ISI publications, INFN Affiliation: 2020- today

- **QUANTEP INFN Call**
  PhD Thesis: *Fabrication and characterization of telecom quantum emitters in silicon for quantum technology applications.* (Awarded with IEEE Gatti Manfredi Award 2025)
- **ROUGE:** Advanced irradiation strategies, collimation masks
- **AURORA:** quantum sensing protocols
- **PNRR NQSTI, Spoke4**
  Controlled fabrication of SPs

SPARQS outcomes

Supporting results

Photon-based quantum technologies:

- Turnkey approach compatible with INFN **SOI technology**.
- Native **Telecom qubits** addressing current scientific efforts in photon frequency conversion.
- Single photon sources **compatible with CMOS-based cryogenic control** of quantum circuits

Other technological and interdisciplinary research:

- Expanding the capabilities of the ion implantation facility

CSN5 funded experiments:

- **QUERIS:** Single-photon sources for Quantum enhanced recognition

Research group

Who?

Research Unit 1: **INFN-TO @ Physics Department**
Coordinator: **Greta Andrini, FTE 1**
1 Full professor, 1 Associated professor, 1 Senior Researcher, 1 RTD, 1 Technologist, 2 Post-docs, 2 PhD students, **TOT FTE: 3,4**

- Ion implantation
- Activation and characterization of single emitters
- Electronics for coherent manipulation of spin qubits

Research Unit 2: **INFN-TO @ INRiM**
Coordinator: Gabriele Zanelli, Postdoc INRiM, (Greta Andrini)
2 Director of Research, 1 Senior Researcher, 1 PhD student, **TOT FTE: 1**

- Optical characterization of telecom color centers
- Dedicated ODMR protocols
- IR Single photon detectors (SNSPDs)

Project Timeline

M12:

- **Activation** of **micrometer-sized** regularly spaced **spots** with G quantum emitters
- **PL signal modulation** under **microwave excitation** (laser optical power, microwave intensities, additional external fields)

M24:

- **Commissioning** of the upgraded irradiation chamber.
- Identification of the **optimal emitters g(0) < 0.5 and** ODMR manipulation of **single spins**

WP	Task description	Timeline (months)
		2026 (Q1 Q2 Q3 Q4) 2027 (Q5 Q6 Q7 Q8)
1.1	Processing parameters evaluation in SOI	
1.2	Diffraction-limited laser activation	
1.3	Ion implantation and in-situ laser activation	
2.1	Cryogenic ODMR protocol development	
2.2	Microwave antenna design optimization	
2.3	ODMR on single spins	

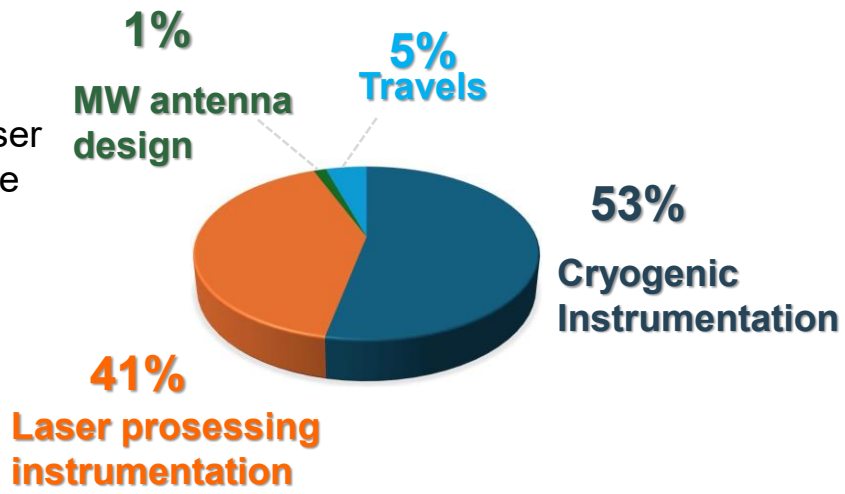
Project budget

1° year (60.5 k€):

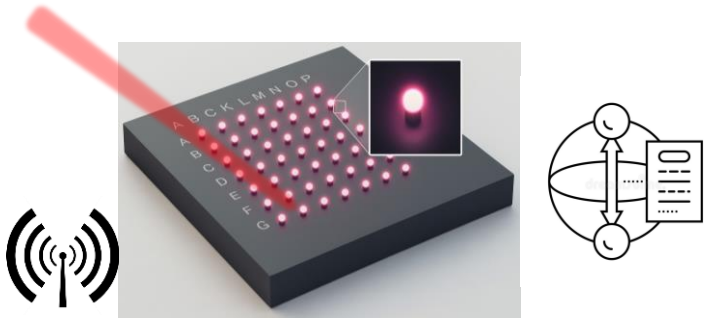
- Acquisition of the high-energy ns-pulsed laser
- Acquisition of the custom reflective objective
- Realization of the dedicated optical path

2nd year (48.5 k€) :

- Acquisition of the Stirling cryocooler
- Acquisition of the dedicated thermal strap
- Design of the MW antenna



Description	Year	Cost (€)
High-vacuum compatible 100x Objective	1	12k
ns Q-switched high energy laser	1	15k
Optics and Optomechanics	1	31k
Travels expenses	1	2.5k
Stirling Cryocooler with Active Vibration Cancellation	2	42k
Thermal strap in OFHC Cu + customization	2	2.5k
Comsol Multiphysics RF module	2	1.5k
Travels expenses	2	2.5k



Thank you for your attention!

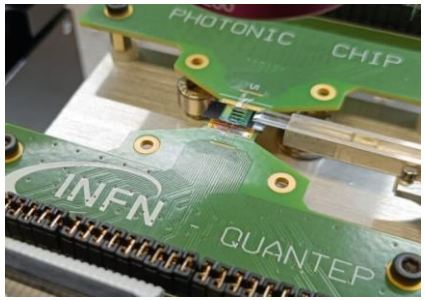
Backup slides

Photonic quantum technologies @ INFN

INFN

2018-2028: **€1 billion for the Quantum Flagship**

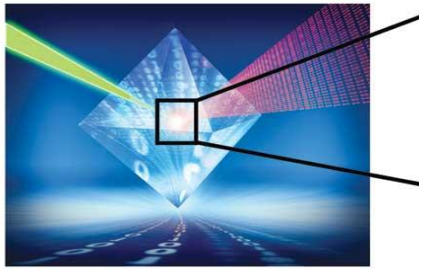
2023-2026: **Piano Nazionale di Ripresa e Resilienza (PNRR), 116 M€**

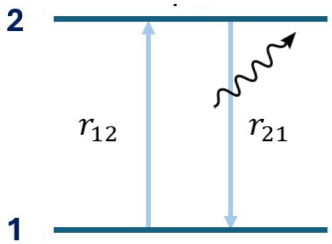


2021-2025: Quantum-oriented funded projects: INFN QUANTEP National Call

Photons as carriers of quantum information:

- ✓ **Store the information** of "stationary qubits."
- ✓ Transfer quantum info over **long distances**
- ✓ Easily detected with **single-photon detectors**

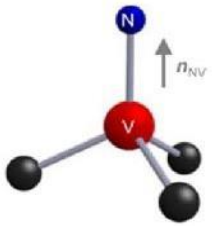


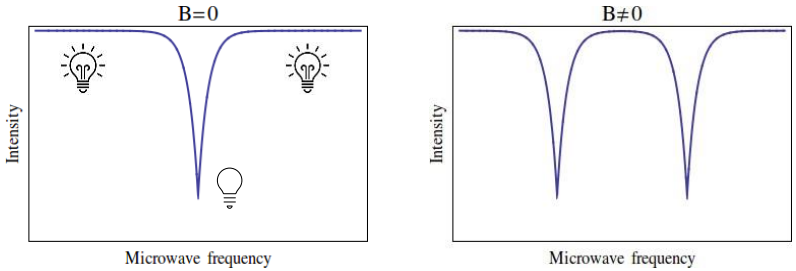


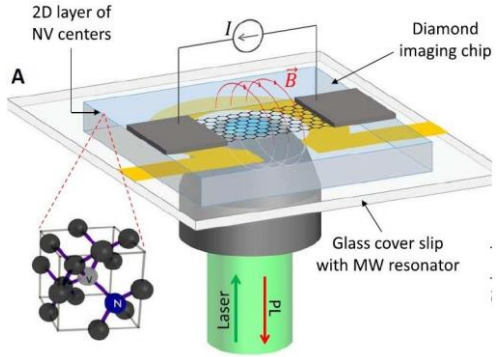
Point defects in solids can provide:

- ✓ **On-demand** single-photon sources
- ✓ A unique **spin-photon interface**

Diamond-based quantum optics





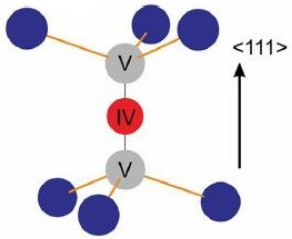


Unique spin properties → quantum sensing: the spin state transition can be coherently driven by microwaves.

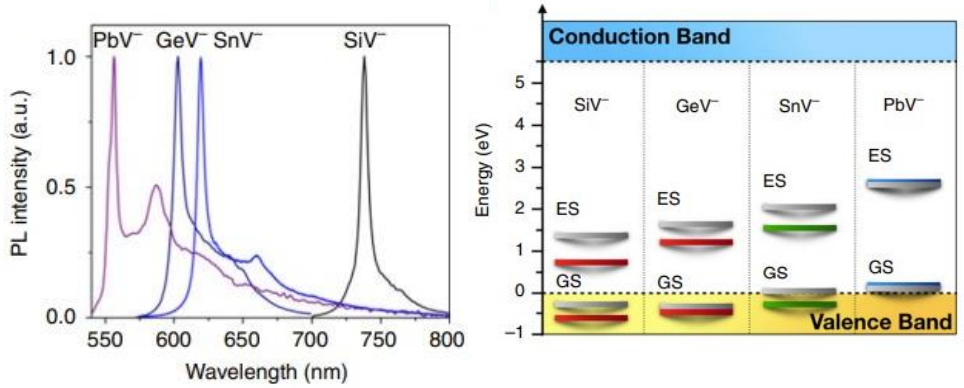
Science Advances 2017, 3 e1602429 Melbourne

Current imaging in 2D materials

Group IV elements for scalable quantum photonic devices



- ns excited state lifetime
- strong emission into the ZPL
- high-fidelity spin control

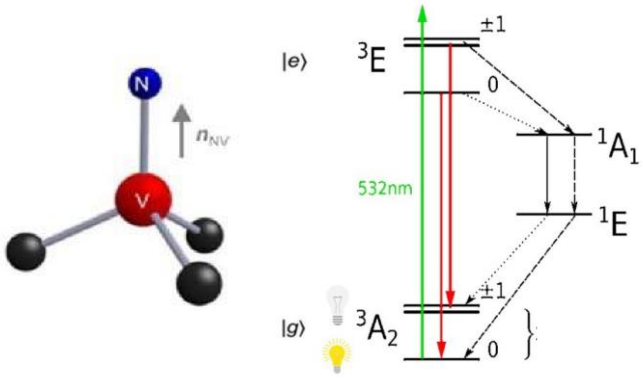


Bradac, C. et al., Nature Communications volume 10, Article number: 5625 (2019)

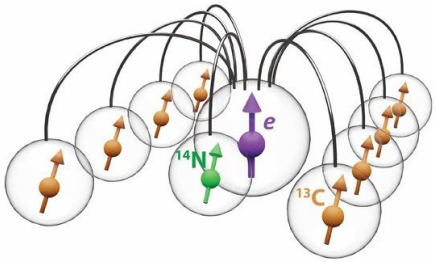
Diamond-based quantum optics

Diamond for **laboratory-scale** quantum memories and quantum networks

The Nitrogen-Vacancy (**NV**) center

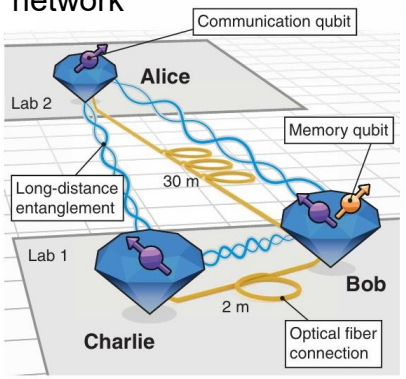


10-qubit quantum register



C. E. Bradley et al. **Phys. Rev. X** 9, 031045 (2019)

3-node remote quantum network



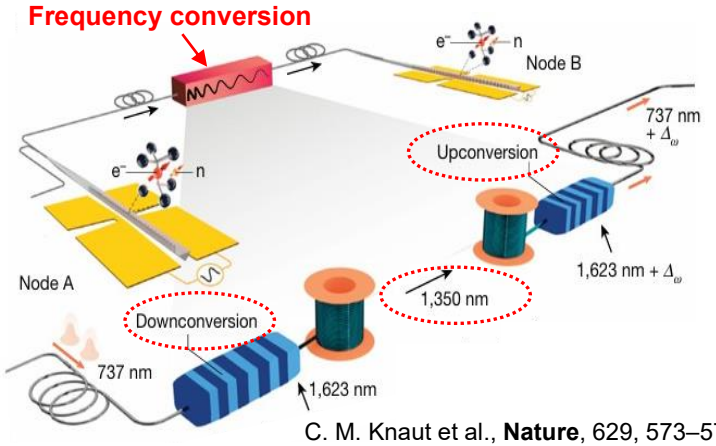
Pompili et al., **Science** 372, 259–264 (2021)

Unique spin properties → quantum information and sensing: microwave-driven coherent control.

👎 **Lack of technology** for wafer-scale diamond photonic integrated circuits

👎 **Frequency conversion** required for low-loss optical components

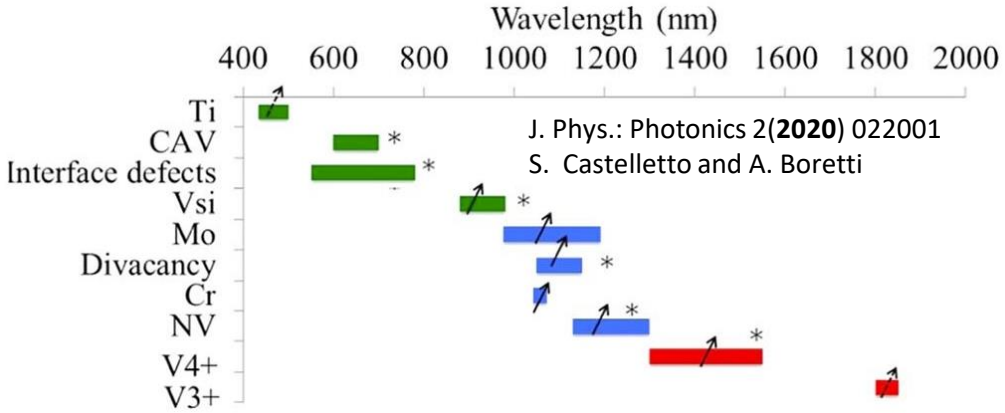
2-node quantum network



C. M. Knaut et al., **Nature**, 629, 573–578 (2024)

SiC-based quantum optics

☑ **Color centers from UV to IR, even at Telecom wavelengths**



J. Phys.: Photonics 2(**2020**) 022001
S. Castelletto and A. Boretti

- Energy gap: 3.3 eV
- Mature production protocols
- High optical nonlinearity
- Suitability for nanofabrication

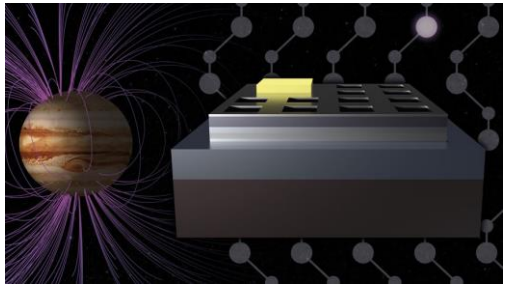
☑ **CMOS Compatible Platform for Harsh-Environment Electronics**

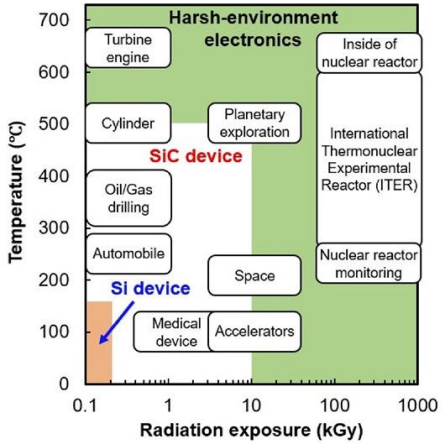
H. Wang et al., Materials Science in Semiconductor Processing 178 (2024)

SiCMag (NASA project) to sense the magnetic fields of planetary bodies:

→ Vector magnetometer for space application

https://microdevices.jpl.nasa.gov/capabilities/nano-and-micro-systems/sic-magnetometer/

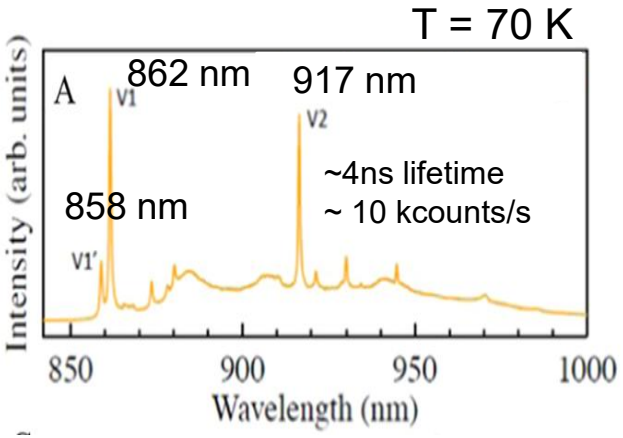




SiC-based quantum optics

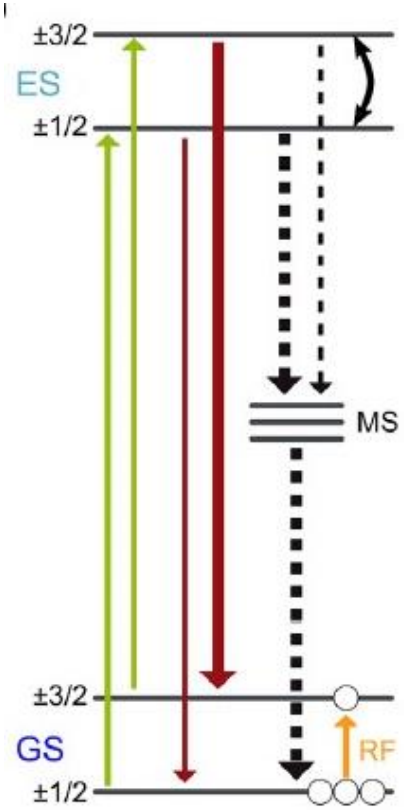
The 4H-SiC crystal structure results in 2 inequivalent lattice sites for the V_{Si} center.

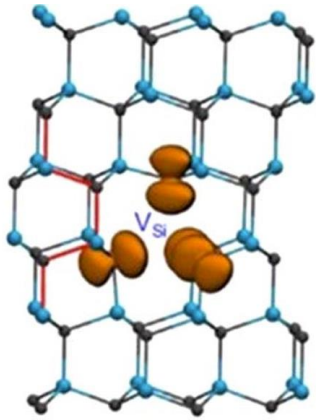
J. Phys.: Photonics 2(**2020**) 022001
S. Castelletto and A. Boretti



- V1, V2 ODMR active
- ms-long spin coherence time
- T_2^* = 30 μs
- High inizialization fidelity

S. A. Tarasenko et al., Phys. Status Solidi B 2017, 00, 1700258



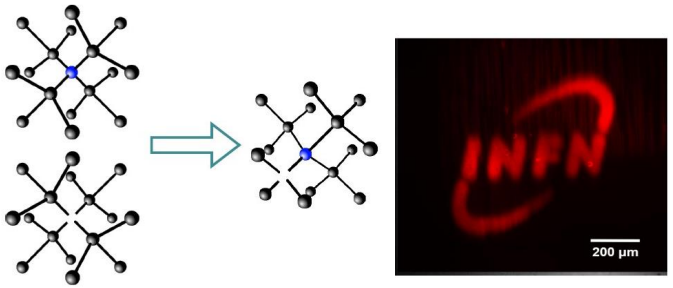


- ☑ Suitability for quantum information applications
- ☑ Nanotesla magnetometry (4.0 nT/√Hz)

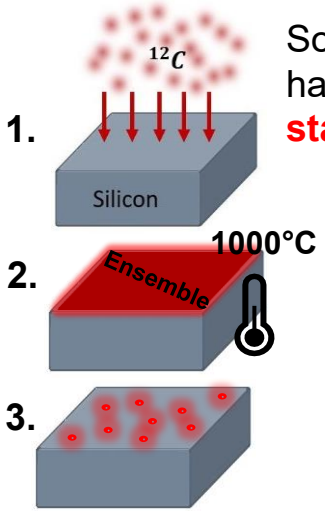
Creation of silicon quantum emitters

State of the art

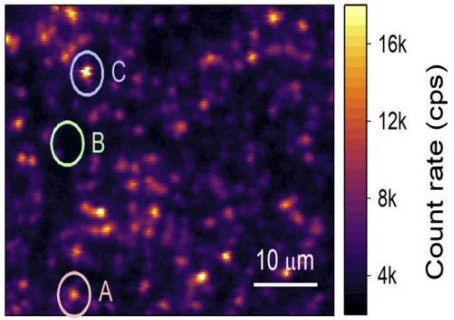
General rule: Ion Implantation + thermal treatment



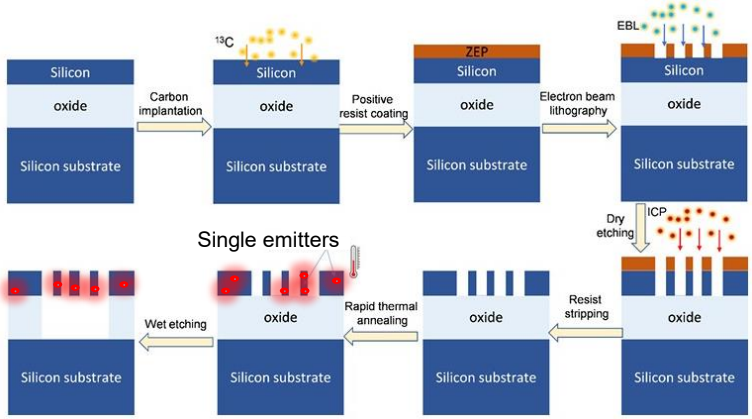
Ion Implantation at LNL INFN



So far, isolating single emitters has mostly relied on a **purely statistical approach**



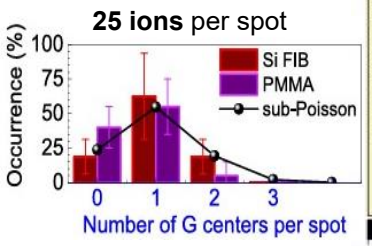
Multi-step fabrication processes to integrate **randomly distributed** emitters into photonic

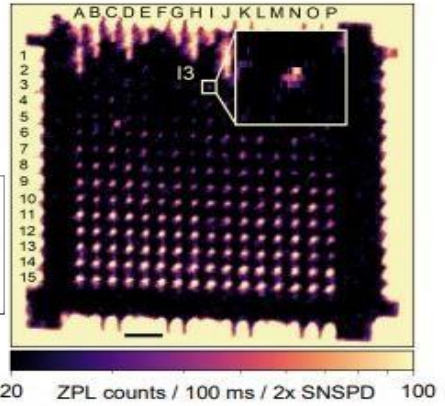


Y. Baron et al., Nat. Comm. 14, 3321 (2023), supplementary info

Focus Ion beam technology to enable the **controlled placement** (~100 nm) of single emitters.

Still probabilistic!





M. Hollenbach et al., Nat. Comm.13, Article number: 7683 (2022)

Research group composition

Who?

Research Unit 1: **INFN-TO @ Physics Department**

Name	Position	Institution	FTE
Greta Andrini	Post-doctoral fellow	INFN Torino	1.0
Matteo Ziino	PhD student	Physics Dep., UniTO (INFN associated)	0.6
Vanna Pugliese	PhD student	Physics Dep., UniTO (INFN associated)	0.5
Emilio Corte	Post-doctoral fellow	Physics Dep., UniTO (INFN associated)	0.5
Alberto Bortone	Technologist	INFN Torino	0.3
Ettore Vittone	Full professor	Physics Dep., UniTO (INFN associated)	0.2
Sviatoslav Ditalia Tchernij	RTD	Physics Dep., UniTO (INFN associated)	0.1
Jacopo Forneris	Associated professor	Physics Dep., UniTO (INFN associated)	0.1
Manuel Rolo	Senior researcher	INFN Torino	0.1

Research Unit 2: **INFN-TO @ INRiM**

Name	Position	Institution	FTE
Gabriele Zanelli	PhD student	Physics Dep., UniTO (INFN associated)	0.3
Ivo Pietro Degiovanni	Director of Research	INRiM (INFN associated)	0.3
Ivano Ruo Berchera	Senior researcher	INRiM (INFN associated)	0.2
Marco Genovese	Director of Research	INRiM (INFN associated)	0.2

Risks and mitigation

Delay in delivery of the **Q-switched** laser or related **optical elements**	• **Other companies** • Objective **without custom design** for off-axis geometry
Unpredictable malfunction or interaction effects of the **Q-switched laser** for localized in-situ activation.	• Proof-of-concept experiment with **CW low-power lasers** and emitters in diamond
Delay in the **Stirling Cryocooler** or the thermal strap	• It is still possible to perform the controlled fabrication of the emitters • **Ex-situ** cryogenic optical **characterization**
Unsuccessful ODMR measurements: Poor ODMR signal contrast	• **Silicon carbide** as **an alternative** solid-state platform with consolidated ODMR and laser-annealing protocols
Unpredictable malfunctioning of the INFN-TO **ion implanter**	• The PI benefit from **access** to the **LNL AN2000 facility** and beamtime for **FIB technology at the HZDR** center and **FBK institute**. • **Ex-situ** cryogenic characterization

Future perspectives

- Realization of **embedded SPs** by combining the end-to-end experimental methodology developed in the SPARQS project with dedicated **photonic structures** (QUANTEP know-how).

- **Investigation** of alternative fabrication protocols based on **cryogenic implantations** for accessing **higher creation efficiency** of the quantum emitters.

- **Synergistic integration** of the end-to-end experimental workflow with a dedicated **single-ion-counting protocol**, ensuring precise control of process uniformity and efficiency.

Consolidated strategies at INFN-TO

Single-photon sensitive cryogenic confocal microscopy to address SPs

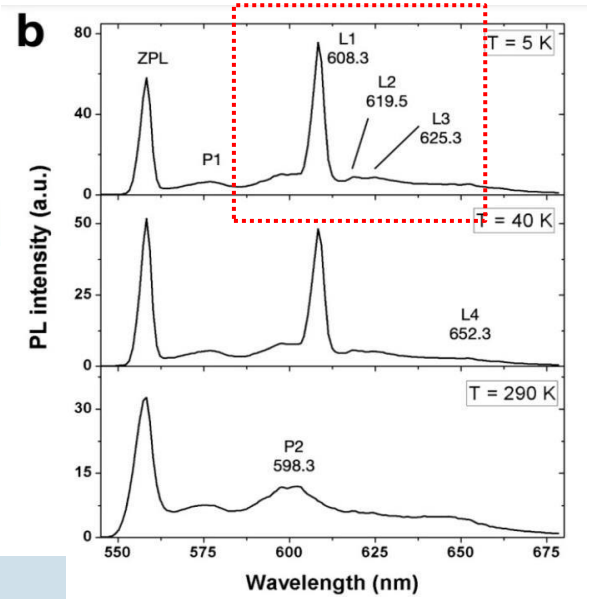
Unprecedented insights into the MgV optical activity!

pubs.acs.org/journal/apchd5 Article

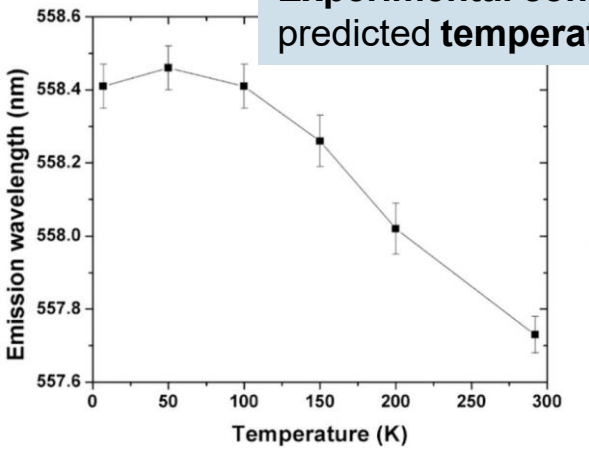
Magnesium-Vacancy Optical Centers in Diamond

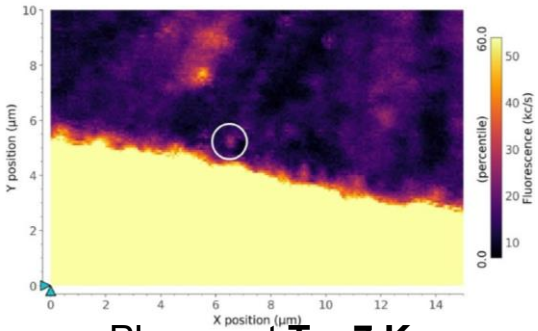
Emilio Corte,[▽] Greta Andrini,[▽] Elena Nieto Hernández, Vanna Pugliese, Ângelo Costa, Goele Magchiels, Janni Moens, Shandirai Malven Tunhuma, Renan Villarreal, Lino M. C. Pereira, André Vantomme, João Guilherme Correia, Ettore Bernardi, Paolo Traina, Ivo Pietro Degiovanni, Ekaterina Moreva, Marco Genovese, Sviatoslav Ditalia Tchernij, Paolo Olivero, Ulrich Wahl,* and Jacopo Forneris*

Cite This: https://doi.org/10.1021/acsphotonics.2c01130 Read Online



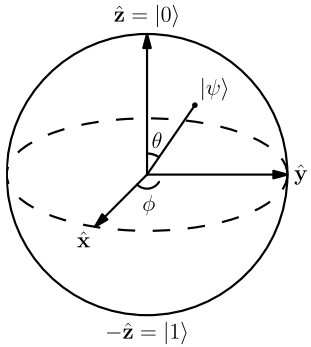
Experimental confirmation of the system's predicted **temperature tunability**





PL map at **T = 7 K**

Future qubit?



Consolidated strategies at INFN-TO

Investigation of the competing processes involved in forming defect complexes

Conferences > 2023 IEEE Photonics Society S...

Study of W centers formation in silicon upon ion implantation and rapid thermal annealing

Publisher: IEEE | Cite This | PDF

G. Andrini ; G. Zanelli ; S. Ditalia Tchernij ; E. Corte ; E. Nieto Hernandez ; A. Verna **All Authors**

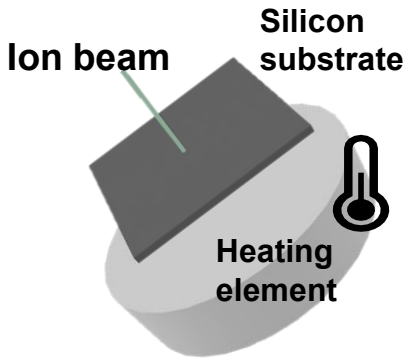
hν photonics MDPI

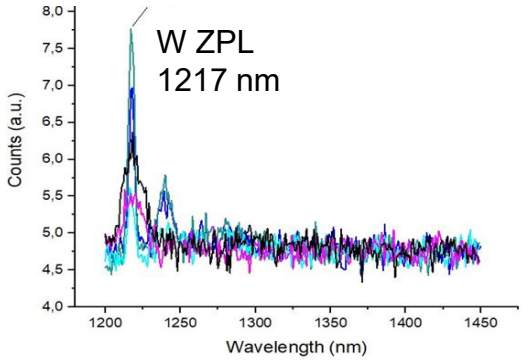
Review

Solid-State Color Centers for Single-Photon Generation

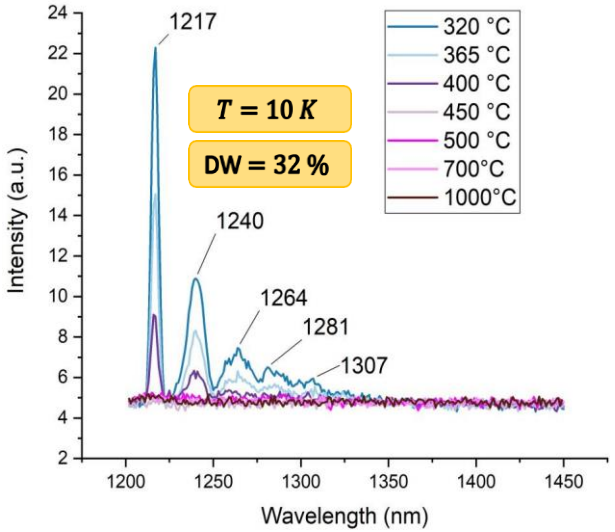
Greta Andrini [1,2,*], Francesco Amanti [3], Fabrizio Armani [4], Vittorio Bellani [5,6], Vincenzo Bonaiuto [7,8], Simone Cammarata [3,9], Matteo Campostrini [10], Thu Ha Dao [7,8], Fabio De Matteis [7,8], Valeria Demontis [6,11], Giovanni Di Giuseppe [12,13], Sviatoslav Ditalia Tchernij [2,14], Simone Donati [3,15], Andrea Fontana [6], Jacopo Forneris [2,14], Roberto Francini [7,8], Luca Frontini [4,16], Roberto Gunnella [12,13], Simone Iadanza [17], Ali Emre Kaplan [18], Cosimo Lacava [6,19], Valentino Liberali [4,16], Francesco Marzioni [12,13,20], Elena Nieto Hernández [2,14], Elena Pedreschi [3], Paolo Piergentili [12,13], Domenic Prete [6], Paolo Prosposito [7,8], Valentino Rigato [10], Carlo Roncolato [10], Francesco Rossella [6,21], Andrea Salamon [8], Matteo Salvato [8,22], Fausto Sargeni [8,23], Jafar Shojaii [24], Franco Spinella [3], Alberto Stabile [4,16], Alessandra Toncelli [3,15], Gabriella Trucco [6,25] and Valerio Vitali [6,19]

«Hot» implantation and dual beam implantation

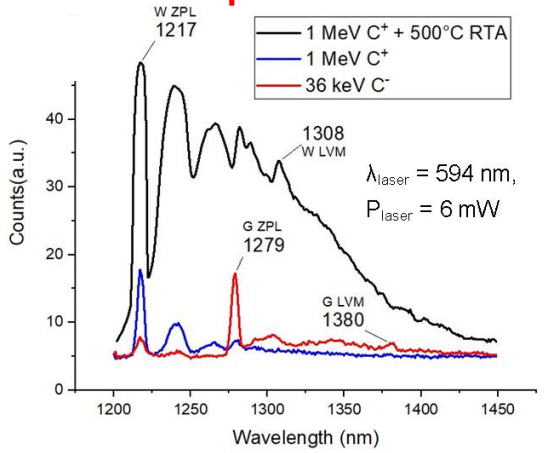




keV implantation



MeV implantation



Consolidated strategies at INFN-TO

Supporting results

Investigation of **formation dynamics** and defect **engineering**

communications materials | Article

Activation of telecom emitters in silicon upon ion implantation and ns pulsed laser annealing

Greta Andrini[1,2], Gabriele Zanelli[3,4], Sviatoslav Ditalia Tchernij[1,3,4], Emilio Corte[3], Elena Nieto Hernández[3], Alessio Verna[5], Matteo Cocuzza[5], Ettore Bernardi[4], Salvatore Virzì[4], Paolo Traina[4], Ivo P. Degiovanni[1,4], Marco Genovese[1,4], Paolo Olivero[1,3,4] & Jacopo Forneris[1,3,4]

scientific reports

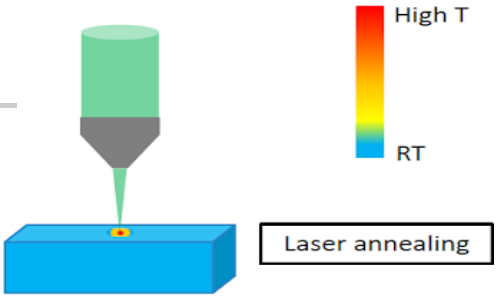
Article | Open access | Published: 12 November 2025

Formation of luminescent defects in 4H-SiC upon ion irradiation and ns laser annealing

Gabriele Zanelli, Igor Fontan, Greta Andrini, Emilio Corte, Elena Nieto Hernández, Veronica Varzi, Nour-Hanne Amine, Sofia Sturari, Elisa Redolfi, Valentino Rigato, Matteo Campostrini, Paolo Traina, Federico Picollo, Marco Genovese, Sviatoslav Ditalia Tchernij & Jacopo Forneris

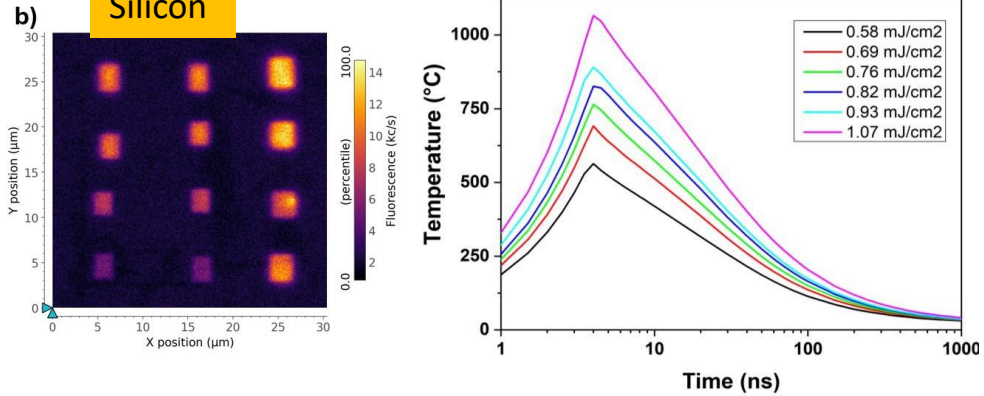
Scientific Reports **15**, Article number: 39640 (2025) | Cite this article

Controlled and **localized ex-situ** activation of G centers.

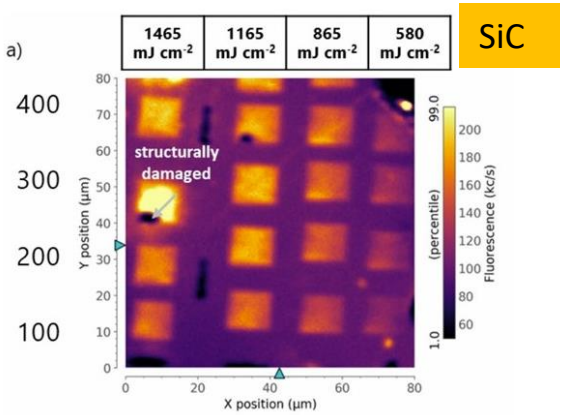


Thermal budget below the melting point!

Silicon



SiC



Consolidated strategies at INFN

Supporting results

Ion Beam Induced Charge for single ion counting

Vacuum

journal homepage: www.elsevier.com/locate/vacuum

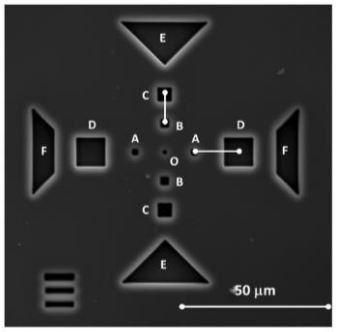
Ramo theorem

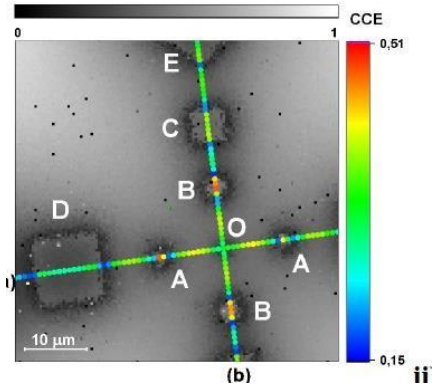
An ion beam spot size monitor based on a nano-machined Si photodiode probed by means of the ion beam induced charge technique

G. Andrini [c], E. Nieto Hernández [a], G. Provatas [b], M. Brajkovic [b], A. Crnjac [b], S. Ditalia Tchernij [a], J. Forneris [a,*], V. Rigato [d], M. Campostrini [d], Z. Siketic [b], M. Jaksic [b], E. Vittone [a]

[a] Physics Dept., University of Torino, and Istituto Nazionale di Fisica Nucleare Sez. Torino, Torino, Italy
[b] Laboratory for Ion Beam Interactions, Ruđer Bošković Institute, Zagreb, Croatia
[c] Dipartimento di Elettronica e Telecomunicazioni, Politecnico di Torino, and Istituto Nazionale di Fisica Nucleare Sez. Torino, Torino, Italy
[d] Istituto Nazionale di Fisica Nucleare, Laboratori Nazionali di Legnaro, Italy

μm or nm beam → **map the device response to the single ion impact**!





SEM picture (left) **and Charge Collection Efficiency map** (right) of a nano-machined Si photodiode

- ✓ **Single ion** sensitivity
- ✓ **Same wafer** to be processed by individual ions
- ✓ **Probing structure** directly on the same wafer

Proof of concept experiment: **in-situ monitoring of a MeV ion micro-beam resolution** exploiting the spatial correlation between the Charge Collection Efficiency (CCE) and probing micro-structures